**SIMATS ENGINEERING**
**ASSESSMENT TOOL 3**

**COURSE NAME: Fundamentals of Data Science** **COURSE CODE: DSA04**

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**COURSE OUTCOME COVERED**

***CO1:*** *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

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**QUESTIONS: 05** **Total Marks:100**

Annexure A

**Exploratory Data Analysis - Based Practical Assessment**

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***Code of Conduct:***

*I SIVA SUBRAMANIYAM B/192572089 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



## Annexure A

## Exploratory Data Analysis-Based Practical Assessment

### 1. Education Analytics Dataset

**Question 1**

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

| Student_ID | Attendance | Study_Hours | Internal_Marks | Result |
|---|---|---|---|---|
| S01 | 92 | 5 | 45 | Pass |
| S02 | 55 | 2 | 24 | Fail |
| S03 | 78 | 4 | 38 | Pass |
| S04 | 60 | 3 | 28 | Fail |
| S05 | 88 | 5 | 42 | Pass |
| S06 | 47 | 1 | 20 | Fail |
| S07 | 95 | 6 | 48 | Pass |
| S08 | 72 | 3 | 35 | Pass |
| S09 | 50 | 2 | 22 | Fail |
| S10 | 83 | 4 | 40 | Pass |
| S11 | 66 | 3 | 30 | Fail |
| S12 | 90 | 5 | 44 | Pass |
| S13 | 58 | 2 | 25 | Fail |
| S14 | 76 | 4 | 36 | Pass |
| S15 | 62 | 3 | 29 | Fail |
| S16 | 85 | 5 | 41 | Pass |
| S17 | 45 | 1 | 18 | Fail |
| S18 | 80 | 4 | 39 | Pass |
| S19 | 68 | 3 | 32 | Pass |
| S20 | 53 | 2 | 23 | Fail |

### 2. Healthcare Analytics Dataset

**Question 2**

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

| Patient_ID | Age | Sugar_Level | BP | BMI | Disease_Status |
|---|---|---|---|---|---|
| P01 | 45 | 145 | 130 | 28 | Yes |
| P02 | 32 | 98 | 118 | 23 | No |
| P03 | 55 | 180 | 145 | 31 | Yes |
| P04 | 28 | 90 | 110 | 22 | No |
| P05 | 60 | 200 | 150 | 34 | Yes |
| P06 | 40 | 120 | 125 | 26 | No |
| P07 | 52 | 165 | 140 | 30 | Yes |
| P08 | 35 | 105 | 120 | 24 | No |
| P09 | 48 | 155 | 135 | 29 | Yes |
| P10 | 30 | 95 | 115 | 23 | No |
| P11 | 62 | 210 | 155 | 35 | Yes |
| P12 | 38 | 115 | 122 | 25 | No |

| | | | | | |
|---|---|---|---|---|---|
| P13 | 50 | 170 | 142 | 30 | Yes |
| P14 | 27 | 88 | 108 | 21 | No |
| P15 | 58 | 190 | 148 | 33 | Yes |
| P16 | 42 | 130 | 128 | 27 | No |
| P17 | 46 | 150 | 132 | 28 | Yes |
| P18 | 34 | 100 | 116 | 24 | No |
| P19 | 53 | 175 | 144 | 31 | Yes |
| P20 | 29 | 92 | 112 | 22 | No |

**3. Retail Sales Dataset**
**Question 3**
A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

| Product_ID | Category | Price | Quantity_Sold | Discount | Revenue |
|---|---|---|---|---|---|
| PR01 | Grocery | 50 | 20 | 5 | 1000 |
| PR02 | Snacks | 30 | 35 | 3 | 1050 |
| PR03 | Dairy | 45 | 18 | 2 | 810 |
| PR04 | Fruits | 60 | 25 | 5 | 1500 |
| PR05 | Vegetables | 40 | 30 | 4 | 1200 |
| PR06 | Grocery | 80 | 15 | 6 | 1200 |
| PR07 | Snacks | 25 | 40 | 3 | 1000 |
| PR08 | Dairy | 55 | 16 | 2 | 880 |
| PR09 | Fruits | 70 | 22 | 5 | 1540 |
| PR10 | Vegetables | 35 | 28 | 4 | 980 |
| PR11 | Grocery | 90 | 12 | 6 | 1080 |
| PR12 | Snacks | 20 | 45 | 2 | 900 |
| PR13 | Dairy | 65 | 14 | 3 | 910 |
| PR14 | Fruits | 75 | 20 | 5 | 1500 |
| PR15 | Vegetables | 45 | 26 | 4 | 1170 |
| PR16 | Grocery | 100 | 10 | 7 | 1000 |
| PR17 | Snacks | 35 | 32 | 3 | 1120 |
| PR18 | Dairy | 50 | 19 | 2 | 950 |
| PR19 | Fruits | 80 | 18 | 6 | 1440 |
| PR20 | Vegetables | 30 | 35 | 3 | 1050 |

**4. Banking Loan Approval Dataset**
**Question 4**
A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

| Customer_ID | Income | Credit_Score | Loan_Amount | Employment_Type | Loan_Status |
|---|---|---|---|---|---|
| C01 | 45000 | 720 | 200000 | Salaried | Approved |
| C02 | 25000 | 580 | 150000 | Self-employed | Rejected |
| C03 | 60000 | 750 | 300000 | Salaried | Approved |
| C04 | 22000 | 560 | 120000 | Unemployed | Rejected |
| C05 | 50000 | 710 | 250000 | Salaried | Approved |
| C06 | 30000 | 600 | 180000 | Self-employed | Rejected |

| | | | | | |
|---|---|---|---|---|---|
| C07 | 70000 | 780 | 350000 | Salaried | Approved |
| C08 | 28000 | 590 | 160000 | Self-employed | Rejected |
| C09 | 55000 | 730 | 270000 | Salaried | Approved |
| C10 | 24000 | 570 | 130000 | Unemployed | Rejected |
| C11 | 65000 | 760 | 320000 | Salaried | Approved |
| C12 | 35000 | 620 | 200000 | Self-employed | Rejected |
| C13 | 48000 | 700 | 240000 | Salaried | Approved |
| C14 | 26000 | 585 | 140000 | Unemployed | Rejected |
| C15 | 75000 | 790 | 400000 | Salaried | Approved |
| C16 | 32000 | 610 | 190000 | Self-employed | Rejected |
| C17 | 52000 | 725 | 260000 | Salaried | Approved |
| C18 | 27000 | 575 | 150000 | Self-employed | Rejected |
| C19 | 68000 | 770 | 330000 | Salaried | Approved |
| C20 | 23000 | 555 | 125000 | Unemployed | Rejected |

**5. Agriculture Crop Yield Dataset**

**Question 5**

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

| Farm_ID | Rainfall_mm | Temperature | Fertilizer_kg | Soil_Type | Crop_Yield_kg |
|---|---|---|---|---|---|
| F01 | 850 | 28 | 60 | Loamy | 3200 |
| F02 | 600 | 32 | 45 | Sandy | 2100 |
| F03 | 900 | 27 | 65 | Loamy | 3500 |
| F04 | 500 | 34 | 40 | Sandy | 1800 |
| F05 | 780 | 29 | 55 | Clay | 3000 |
| F06 | 650 | 31 | 48 | Sandy | 2300 |
| F07 | 920 | 26 | 70 | Loamy | 3700 |
| F08 | 720 | 30 | 52 | Clay | 2800 |
| F09 | 550 | 33 | 42 | Sandy | 1900 |
| F10 | 800 | 28 | 58 | Loamy | 3100 |
| F11 | 880 | 27 | 62 | Loamy | 3400 |
| F12 | 620 | 32 | 46 | Sandy | 2200 |
| F13 | 760 | 29 | 54 | Clay | 2950 |
| F14 | 940 | 26 | 72 | Loamy | 3800 |
| F15 | 580 | 33 | 44 | Sandy | 2000 |
| F16 | 820 | 28 | 60 | Clay | 3150 |
| F17 | 700 | 30 | 50 | Clay | 2700 |
| F18 | 960 | 25 | 75 | Loamy | 3900 |
| F19 | 540 | 34 | 41 | Sandy | 1850 |
| F20 | 790 | 29 | 56 | Clay | 3050 |

**Common EDA Tasks:**

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.

6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

## RUBRICS FOR CALCULATION

| Exploratory Data Analysis-Based Practical Assessment | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
| **Dataset Understanding** | 20 | Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity | Explains dataset, features, and variables with adequate clarity | Basic explanation of dataset with limited understanding of variables | Poor understanding of dataset; variables and problem context are unclear |
| **Data Cleaning** | 25 | Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods | Handles most missing values, duplicates, and errors correctly | Performs basic cleaning but misses some important data quality issues | Minimal or incorrect data cleaning; major errors remain in the dataset |
| **Data Analysis and Statistical Summary** | 20 | Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations | Provides relevant summary statistics and basic interpretation | Provides limited statistics with weak explanation | Lacks statistical analysis or gives incorrect interpretation |
| **Data Visualization** | 20 | Uses highly appropriate charts with proper labels, titles, and clear visual interpretation | Uses suitable charts with mostly clear labels and explanation | Uses limited or basic charts with unclear interpretation | Poor or incorrect visualizations; charts do not support analysis |
| **Insight Generation and Reporting** | 15 | Presents strong insights, conclusions, and recommendations based on data evidence | Presents clear findings and conclusions with reasonable support | Provides basic findings but lacks depth or proper justification | Findings are unclear, unsupported, or not related to the data |

**Exploratory Data Analysis (EDA) — Practical Assessment Report**

**Question 1: Education Analytics Dataset**

**1. Dataset Understanding & Metadata**

- **Context**: Identify factors influencing student academic results (Result) using behavioral and continuous assessment indicators.
- **Shape**: 20 rows, 5 columns.
- **Data Dictionary & Types**:
    - Student_ID: Unique student identifier (Categorical / Nominal, object)
    - Attendance: Percentage of classes attended (Numerical / Discrete, int64)
    - Study_Hours: Daily self-study hours (Numerical / Discrete, int64)
    - Internal_Marks: Continuous evaluation score out of 50 (Numerical / Discrete, int64)
    - Result: Academic outcome — Pass or Fail (Target Variable, Categorical / Nominal, object)

**2. Data Cleaning & Integrity Check**

- **Missing Values**: 0 null/missing entries across all 5 features.
- **Duplicates**: 0 duplicate rows; all 20 Student_IDs are unique (S01 to S20).
- **Outlier Detection**: No out-of-range percentages ($0 \leq \text{Attendance} \leq 100$) or irregular values.

### 3. Statistical Summary

| Metric | Attendance (%) | Study Hours | Internal Marks |
|---|---|---|---|
| **Mean** | 69.85 | 3.50 | 32.50 |
| **Median** | 70.00 | 3.50 | 33.50 |
| **Min** | 45.00 | 1.00 | 18.00 |
| **Max** | 95.00 | 6.00 | 48.00 |
| **Std Dev** | 16.74 | 1.36 | 8.88 |

**Class-Wise Subgroup Analysis**

| Class | Count | Mean Attendance | Mean Study Hours | Mean Internal Marks |
|---|---|---|---|---|
| **Pass** | 11 (55%) | 82.64% | 4.45 hrs | 39.09 |
| **Fail** | 9 (45%) | 54.22% | 2.33 hrs | 24.44 |

## 4. Data Visualization Plan & Analysis

- **Histogram**: Distribution of Attendance exhibits a bimodal structure clustering around 50–60% (struggling cohort) and 75–95% (high-performing cohort).
- **Box Plot**: Internal_Marks grouped by Result displays distinct interquartile ranges with no overlap (Fail: 18–30; Pass: 32–48).
- **Scatter Plot**: Attendance vs. Internal_Marks (hue: Result) reveals a direct linear relationship; a threshold line at Attendance $\geq 68\%$ and Internal_Marks $\geq 32$ separates passing from failing students.

## 5. Observations

1. **Attendance Threshold**: Every student who passed had an attendance rate of at least 68% (S19), whereas all students with attendance below 68% failed.
2. **Study Hours Impact**: Passing students dedicate an average of 4.45 hours/day to studying compared to 2.33 hours/day for failing students.
3. **Passing Boundary**: The minimum internal mark recorded for a passing student was 32 (S19), while the highest mark among failing students was 30 (S11).
4. **Strong Multicollinearity**: Higher study hours directly correlate with higher attendance and internal marks.

## 6. Conclusion

Academic success in this cohort is determined by class attendance and daily study hours. Establishing an early warning intervention when attendance falls below 70% or study hours drop under 3 hours/day can help prevent student failure.

**Question 2: Healthcare Analytics Dataset**

**1. Dataset Understanding & Metadata**

- **Context**: Determine clinical indicators associated with chronic disease risk (Disease_Status).
- **Shape**: 20 rows, 6 columns.
- **Data Dictionary & Types**:
    - Patient_ID: Unique patient key (Nominal, object)
    - Age: Patient age in years (Discrete Numerical, int64)
    - Sugar_Level: Fasting blood sugar in mg/dL (Continuous Numerical, int64)
    - BP: Systolic blood pressure in mmHg (Continuous Numerical, int64)
    - BMI: Body Mass Index in $kg/m^2$ (Continuous Numerical, int64)
    - Disease_Status: Binary target indicator (Yes / No) (Nominal, object)

**2. Data Cleaning & Integrity Check**

- **Missing Values**: 0 null records across all parameters.
- **Duplicates**: 0 duplicate entries across all rows.
- **Validity**: All clinical metrics fall within physically plausible ranges.

### 3. Statistical Summary

| Metric | Age (years) | Sugar Level (mg/dL) | BP (mmHg) | BMI |
|---|---|---|---|---|
| **Mean** | 43.10 | 143.40 | 130.40 | 27.05 |
| **Median** | 43.50 | 147.50 | 129.00 | 27.00 |
| **Min** | 27.00 | 88.00 | 108.00 | 21.00 |
| **Max** | 62.00 | 210.00 | 155.00 | 35.00 |
| **Std Dev** | 11.75 | 39.81 | 14.50 | 4.38 |

**Cohort Segmentation by Disease Status**

| Status | Count | Mean Age | Mean Sugar Level | Mean BP | Mean BMI |
|---|---|---|---|---|---|
| **Yes** | 10 (50%) | 53.40 | 173.50 mg/dL | 141.90 mmHg | 31.10 |
| **No** | 10 (50%) | 32.80 | 113.30 mg/dL | 118.90 mmHg | 23.00 |

## 4. Data Visualization Plan & Analysis

- **Box Plot**: Sugar_Level by Disease_Status indicates clear class separation: healthy individuals have sugar levels below 135 mg/dL, whereas individuals with positive disease status measure at or above 145 mg/dL.
- **Scatter Plot**: Age vs. BP shows a positive trend; individuals over age 45 consistently display stage-1 or stage-2 hypertension ($BP \geq 130$ mmHg).
- **Bar Chart**: Mean BMI comparison illustrates that diseased individuals fall into the Obese category (BMI $\geq 30$), while non-diseased individuals average a normal BMI (23.0).

## 5. Observations

1. **Complete Class Balance**: The dataset has an exact 50:50 distribution (10 Yes, 10 No).
2. **Age Threshold**: All patients diagnosed with the disease are 45 years or older (P01 at 45 is the youngest).
3. **Glycemic Biomarker**: Sugar levels in the positive group range from 145 to 210 mg/dL, indicating chronic hyperglycemia.
4. **Cardiovascular Risk Co-occurrence**: Patients with elevated blood pressure ($BP \geq 130$ mmHg) also show elevated BMI ($\geq 28$).

## 6. Conclusion

Age ($\geq 45$), hyperglycemia ($\geq 145$ mg/dL), and elevated BMI ($\geq 28$) serve as strong indicators of disease status, highlighting the value of targeted screening programs for patients aged 40 and older.

**Question 3: Retail Sales Dataset**

**1. Dataset Understanding & Metadata**

- **Context**: Identify top-selling items, category performance, and pricing/discount drivers of revenue.
- **Shape**: 20 rows, 6 columns.
- **Data Dictionary & Types**:
    - Product_ID: Product SKU identifier (Nominal, object)
    - Category: Merchandise classification (Nominal, object)
    - Price: Unit price in INR/USD (Continuous Numerical, int64)
    - Quantity_Sold: Volume of units sold (Discrete Numerical, int64)
    - Discount: Percentage discount applied (Discrete Numerical, int64)
    - Revenue: Total sales revenue generated (Price × Quantity) (Numerical, int64)

**2. Data Cleaning & Integrity Check**

- **Missing Values**: None detected.
- **Formula Verification**: Verified that Revenue = Price × Quantity_Sold holds across all rows (discounts reflect list promotions without reducing reported gross earnings).
- **Duplicates**: 20 unique product entries.

**3. Statistical Summary**

| Metric | Price | Quantity Sold | Discount (%) | Revenue |
|---|---|---|---|---|
| **Mean** | 53.75 | 24.00 | 3.80 | 1,147.00 |

| Metric | Price | Quantity Sold | Discount (%) | Revenue |
|---|---|---|---|---|
| **Median** | 50.00 | 21.00 | 4.00 | 1,065.00 |
| **Min** | 20.00 | 10.00 | 2.00 | 810.00 |
| **Max** | 100.00 | 45.00 | 7.00 | 1,540.00 |
| **Std Dev** | 22.88 | 9.77 | 1.61 | 224.24 |

**Breakdown by Product Category**

| Category | Count | Avg Price | Total Units Sold | Total Revenue | Avg Revenue/Product |
|---|---|---|---|---|---|
| **Fruits** | 4 | 71.25 | 85 | 5,980 | 1,495.00 |
| **Vegetables** | 4 | 37.50 | 119 | 4,400 | 1,100.00 |
| **Grocery** | 4 | 80.00 | 57 | 4,280 | 1,070.00 |
| **Snacks** | 4 | 27.50 | 152 | 4,070 | 1,017.50 |

| Category | Count | Avg Price | Total Units Sold | Total Revenue | Avg Revenue/Product |
|---|---|---|---|---|---|
| **Dairy** | 4 | 53.75 | 67 | 3,550 | 887.50 |

**4. Data Visualization Plan & Analysis**

- **Bar Chart**: Total Revenue by Category highlights Fruits as the top revenue driver ($5,980) and Dairy as the lowest ($3,550).
- **Scatter Plot**: Price vs. Quantity_Sold demonstrates classic price elasticity, showing an inverse relationship where lower-priced goods achieve higher sales volumes.
- **Box Plot**: Discount distribution across categories shows Grocery receiving higher discounts (6–7%) to offset slower inventory turnover.

**5. Observations**

1. **Top Revenue Driver**: Fruits generated the highest revenue ($5,980 total; average $1,495/product), led by PR09 ($1,540).
2. **Top Volume Driver**: Snacks had the highest sales volume (152 units sold), led by PR12 (45 units sold at a low unit price of 20).
3. **Lowest Performer**: Dairy produced the lowest gross revenue ($3,550), driven by low transaction quantities (14–19 units).
4. **Discount Invariance**: Higher discount rates (up to 7% in Grocery) did not yield higher sales volume relative to inherently fast-moving goods like Snacks and Vegetables.

**6. Conclusion**

Revenue generation follows two main patterns: high unit margins (Fruits) and volume-driven throughput (Snacks). Inventory planning should prioritize high-margin fruit stock while maintaining competitive pricing on snacks to drive overall basket size.

**Question 4: Banking Loan Approval Dataset**

**1. Dataset Understanding & Metadata**

- **Context**: Identify applicant financial and credit profiles linked to loan approval decisions (Loan_Status).
- **Shape**: 20 rows, 6 columns.
- **Data Dictionary & Types**:
    - Customer_ID: Unique applicant key (Nominal, object)
    - Income: Monthly gross income (Continuous Numerical, int64)
    - Credit_Score: Credit bureau score (Continuous Numerical, int64)
    - Loan_Amount: Requested loan principal (Continuous Numerical, int64)
    - Employment_Type: Employment classification (Salaried, Self-employed, Unemployed) (Nominal, object)
    - Loan_Status: Credit decision outcome (Approved / Rejected) (Target, object)

**2. Data Cleaning & Integrity Check**

- **Missing Values**: 0 missing values across all columns.
- **Duplicates**: 0 duplicate rows.
- **Consistency Check**: All credit scores fall within standard reporting boundaries (550 to 800).

### 3. Statistical Summary

| Metric | Income | Credit Score | Loan Amount |
|---|---|---|---|
| **Mean** | 43,650.00 | 668.25 | 226,750.00 |
| **Median** | 40,000.00 | 660.00 | 220,000.00 |
| **Min** | 22,000.00 | 555.00 | 120,000.00 |
| **Max** | 75,000.00 | 790.00 | 400,000.00 |
| **Std Dev** | 18,207.21 | 85.34 | 82,345.15 |

**Comparative Profiles by Loan Decision**

| Metric | Approved (n=10) | Rejected (n=10) |
|---|---|---|
| **Mean Income** | 58,800.00 | 28,500.00 |
| **Mean Credit Score** | 744.00 | 592.50 |
| **Mean Loan Amount** | 297,000.00 | 156,500.00 |

| Metric | Approved (n=10) | Rejected (n=10) |
|---|---|---|
| **Employment Distribution** | 100% Salaried (10/10) | 60% Self-Employed (6), 40% Unemployed (4) |

**4. Data Visualization Plan & Analysis**

- **Grouped Bar Chart**: Employment_Type by Loan_Status shows a clear separation: all salaried applicants were approved, while non-salaried applicants were rejected.
- **Scatter Plot**: Income vs. Credit_Score (hue: Loan_Status) identifies a distinct decision boundary separating approved applicants from rejected applicants.
- **Box Plot**: Credit_Score by status shows approved applicants maintain scores above 700, whereas rejected applicants fall below 620.

**5. Observations**

1. **Employment Dependency**: All 10 approved loans belong to Salaried individuals; 0 self-employed or unemployed applicants secured approval.
2. **Credit Score Threshold**: A credit score of 700 serves as a clear cutoff: all approved applicants had scores $\geq 700$, while rejected applicants had scores $\leq 620$.
3. **Income Separation**: Approved applicants earned an average of 58,800 (minimum 45,000), whereas rejected applicants earned an average of 28,500 (maximum 35,000).
4. **Loan Sizing**: Despite requesting higher principal amounts ($200,000–$400,000), salaried applicants were approved based on their income and credit standing.

**6. Conclusion**

Underwriting decisions depend primarily on employment type, income levels ($\geq 45{,}000$), and credit scores ($\geq 700$). Developing tailored credit products with alternative collateral could help serve self-employed applicants who fall outside standard salaried approval criteria.

**Question 5: Agriculture Crop Yield Dataset**

**1. Dataset Understanding & Metadata**

- **Context**: Determine the influence of meteorological conditions, soil composition, and fertilization on crop yield (Crop_Yield_kg).
- **Shape**: 20 rows, 6 columns.
- **Data Dictionary & Types**:
    - Farm_ID: Farm site identifier (Nominal, object)
    - Rainfall_mm: Total seasonal precipitation in mm (Continuous Numerical, int64)
    - Temperature: Ambient temperature in °C (Continuous Numerical, int64)
    - Fertilizer_kg: Fertilizer applied in kg (Continuous Numerical, int64)
    - Soil_Type: Soil classification (Loamy, Clay, Sandy) (Categorical / Nominal, object)
    - Crop_Yield_kg: Harvest yield in kg (Target Variable, Continuous Numerical, int64)

**2. Data Cleaning & Integrity Check**

- **Missing Values**: 0 null entries across all 20 records.
- **Duplicates**: 0 duplicate farm observations.
- **Consistency Check**: All rainfall, fertilizer, and temperature readings align with expected agricultural ranges.

### 3. Statistical Summary

| Metric | Rainfall (mm) | Temperature (°C) | Fertilizer (kg) | Crop Yield (kg) |
|---|---|---|---|---|
| **Mean** | 747.00 | 29.60 | 54.00 | 2,795.00 |
| **Median** | 770.00 | 29.50 | 54.50 | 2,875.00 |
| **Min** | 500.00 | 25.00 | 40.00 | 1,800.00 |
| **Max** | 960.00 | 34.00 | 75.00 | 3,900.00 |
| **Std Dev** | 148.68 | 2.80 | 10.60 | 674.52 |

**Agronomic Soil Profile Aggregation**

| Soil Type | Count | Avg Rainfall (mm) | Avg Temp (°C) | Avg Fertilizer (kg) | Avg Yield (kg) |
|---|---|---|---|---|---|
| **Loamy** | 7 | 892.86 | 26.71 | 66.00 | 3,514.29 |
| **Clay** | 6 | 761.67 | 29.17 | 54.50 | 2,941.67 |
| **Sandy** | 7 | 577.14 | 32.86 | 43.71 | 2,021.43 |

## 4. Data Visualization Plan & Analysis

- **Box Plot**: Crop_Yield_kg across Soil_Type shows clear separation: Loamy soil produces 3,100–3,900 kg, Clay produces 2,700–3,150 kg, and Sandy soil produces 1,800–2,300 kg.
- **Scatter Plot**: Rainfall_mm vs. Crop_Yield_kg displays a strong positive linear relationship; higher precipitation consistently yields larger harvests.
- **Scatter Plot**: Temperature vs. Crop_Yield_kg reveals an inverse correlation, where temperatures above 31°C correspond to reduced yields.

## 5. Observations

1. **Soil Type Productivity**: Loamy soil yields the highest output (average 3,514.29 kg), outperforming Clay (2,941.67 kg) and Sandy soil (2,021.43 kg).
2. **Rainfall Impact**: Yield scales directly with rainfall; farms receiving over 800 mm consistently produce yields above 3,100 kg.
3. **Fertilizer Correlation**: Higher fertilizer application (60–75 kg) strongly aligns with improved yield outcomes in Loamy soils.
4. **Heat Stress**: Temperatures between 31°C and 34°C coincide with lower rainfall and yield drops down to 1,800 kg.

## 6. Conclusion

Crop production peaks under conditions of high rainfall (850–960 mm), moderate temperatures (25–28°C), higher fertilization, and Loamy soil. For farms with Sandy soils or low rainfall, introducing irrigation and soil conditioning can help improve output.

**Python EDA Reference Script**

**Python:**

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
# Set visualization style
sns.set_theme(style="whitegrid")
def perform_full_eda(df, target_col, dataset_title):
    print(f"=== {dataset_title} ===")
    print(f"Shape: {df.shape}")
    print("\nData Types & Null Counts:")
    print(df.info())
    print("\nDescriptive Statistics (Numerical):")
    print(df.describe())
    print("\nMissing Values Count:")
    print(df.isnull().sum())
    print("\nDuplicate Rows:", df.duplicated().sum())
    if df[target_col].dtype == "object":
        print(f"\nTarget Breakdown ({target_col}):")
        print(df[target_col].value_counts(normalize=True) * 100)
# Example execution for Question 1
edu_data = {
    "Attendance": [
        92,
        55,
        78,
        60,
        88,
        47,
        95,
        72,
        50,
        83,
```

```
    66,
    90,
    58,
    76,
    62,
    85,
    45,
    80,
    68,
    53,
],
"Study_Hours": [5, 2, 4, 3, 5, 1, 6, 3, 2, 4, 3, 5, 2, 4, 3, 5, 1, 4, 3, 2],
"Internal_Marks": [
    45,
    24,
    38,
    28,
    42,
    20,
    48,
    35,
    22,
    40,
    30,
    44,
    25,
    36,
    29,
    41,
    18,
    39,
    32,
    23,
],
```

```
    "Result": [
        "Pass",
        "Fail",
        "Pass",
        "Fail",
        "Pass",
        "Fail",
        "Pass",
        "Pass",
        "Fail",
        "Pass",
        "Fail",
        "Pass",
        "Fail",
        "Pass",
        "Fail",
        "Pass",
        "Pass",
        "Fail",
    ],
}
df_edu = pd.DataFrame(edu_data)
perform_full_eda(df_edu, target_col="Result", dataset_title="Education Analytics")
```